

Solution Architecture for Data Quality Monitoring and AI Implementation

College Name: HKBK College of Engineering

Group Members:

- **Name:** Kavya R Naik
CAN ID Number: CAN_33740336
 - **Name:** Md Modassir Ali
CAN ID Number: CAN_33506401
 - **Name:** Dhakshana Moorthy
CAN ID Number: CAN_33836237
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1. Overview of Data Quality Monitoring

Building on phase 1, the solution architecture emphasizes the need to analyze and visualize data patterns to ensure integrity, completeness, and consistency in cloud data warehouses. Visualizations serve as a powerful tool for uncovering data patterns, trends, and anomalies in cloud-based data warehouses. By implementing dynamic and interactive dashboards, stakeholders can easily spot irregularities and address data quality concerns in real-time.

Objectives:

- Analyze and visualize cloud data warehouse trends to uncover potential quality issues.
 - Develop hypotheses for detecting anomalies in schema, patterns, and volume.
 - Establish the architecture for implementing scalable AI-powered solutions.
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2. Data Cleaning and Preparation

2.1. Data Preparation:

- **Handling Missing Data:**
 - Fill missing sales data using median imputation or trend-based forecasting.

- **Outlier Management:**

- Detect extreme sales values using AI models like Isolation Forest or Winsorization.

- **Standardization:**

- Normalize features like sales revenue and product pricing for model compatibility.

- **Dimensionality Reduction:**

- Use PCA to reduce redundancy while retaining critical sales data patterns.

- **AI Model Selection:**

- **Preferred Model:** Isolation Forest for its robustness, scalability, and interpretability.
- Justification based on precision, recall, and real-time performance.

- Example Code:

```
# Convert 'Order Date' to datetime format and handle missing values
data['Order Date'] = pd.to_datetime(data['Order Date'], errors='coerce')
data.fillna(data.median(numeric_only=True), inplace=True)

# Outlier Detection using Isolation Forest
clf = IsolationForest(contamination=0.01, random_state=42)
data['Outlier'] = clf.fit_predict(data[['Sales']])

# Standardization of features
scaler = StandardScaler()
data['Standardized_Sales'] = scaler.fit_transform(data[['Sales']])
```

3. Visualization Tools for Data Monitoring

- Interactive dashboards for real-time data quality visualization using tools like Plotly and Power BI.
- Example:
- Correlation matrices to highlight redundant or conflicting data relationships.
- Time-series plots to monitor data consistency over time.
- Example code

```
# Calculate the correlation matrix for relevant columns
correlation_matrix = data[['Sales', 'Discount', 'Profit']].corr()

# Create a heatmap using Plotly
```

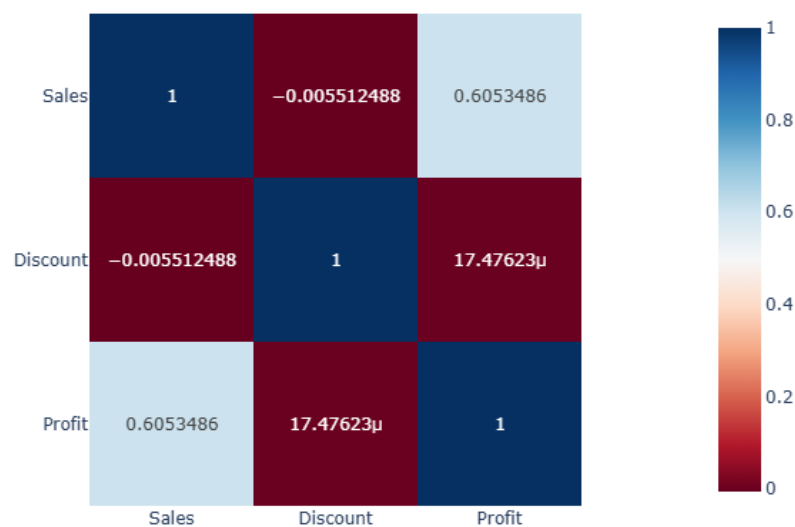
```

fig = px.imshow(correlation_matrix,
                text_auto=True,
                color_continuous_scale='RdBu', # Use a valid Plotly colorscale
                title="Correlation Matrix for Sales, Discount, and Profit")

# Customize layout and display the heatmap
fig.update_layout(title="Correlation Matrix for Sales, Discount, and Profit")
fig.show()

```

Correlation Matrix for Sales, Discount, and Profit



```

# Resample sales data by month and calculate the total sales for each month

```

```

monthly_sales = data['Sales'].resample('M').sum()

```

```

# Plot the monthly sales trend

```

```

plt.figure(figsize=(12, 6))

```

```

plt.plot(monthly_sales, marker='o', linestyle='-', color='blue', label='Monthly Sales')

```

```

plt.title('Monthly Sales Over Time', fontsize=16)

```

```

plt.xlabel('Date', fontsize=12)

```

```

plt.ylabel('Total Sales', fontsize=12)

```

```

plt.grid(True, linestyle='--', alpha=0.6)

```

```

plt.legend(fontsize=12)

```

```

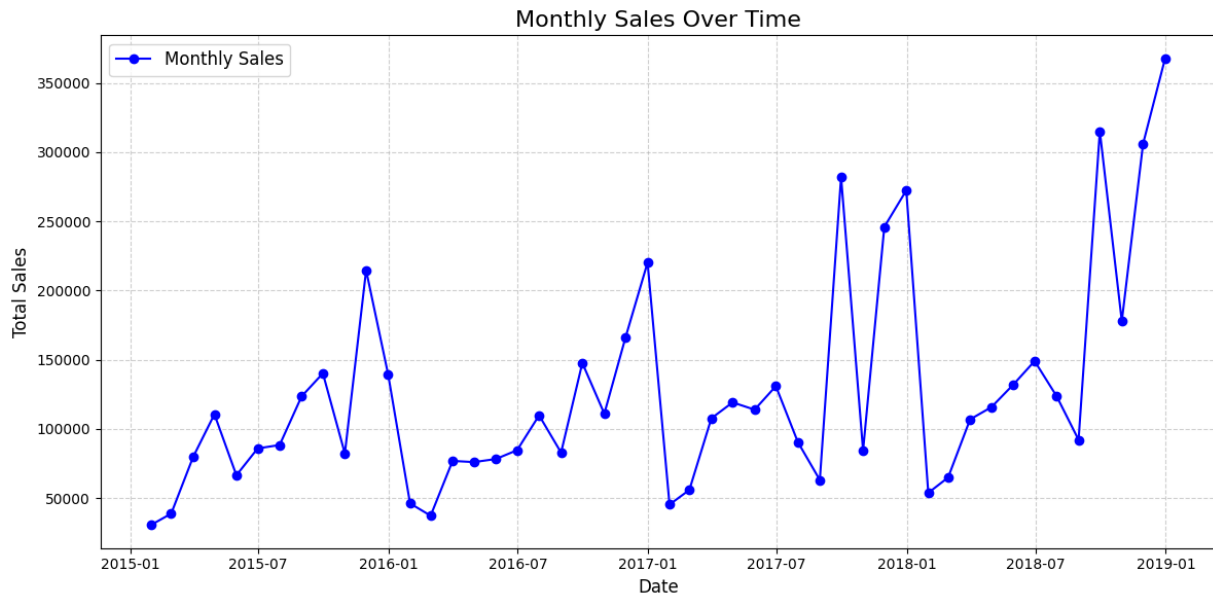
plt.tight_layout()

```

```

plt.show()

```



4. AI Model Selection for Data Quality Monitoring

- **Evaluate algorithms tailored for cloud environments:**
 - **Isolation Forest:**
 - Efficient for high-dimensional and sparse datasets.
 - Excellent for identifying anomalies indicative of data quality issues.
 - Justification: Handles imbalanced data effectively (**Robustness**), Designed for large-scale datasets (**Scalability**).
- **DBSCAN (Density-Based Spatial Clustering of Applications with Noise):**
 - Useful for identifying clusters of anomalies but may struggle with sparse data.
- **One-Class SVM:**
 - Theoretically robust but computationally expensive for real-time monitoring in cloud environments.

5. Data Transformation and Feature Engineering

- Automate feature scaling, encoding, and dimensionality reduction using pipelines in AI platforms like IBM Watson Studio.

6. Feasibility Assessment

- **Define measurable metrics for data quality:**
 - Precision, recall, and ROC-AUC to evaluate AI model effectiveness.
 - Business alignment with real-time data usage scenarios.

7. Conclusion

The integration of advanced visualizations, robust data preparation techniques, and cutting-edge AI models on IBM Cloud provides a comprehensive solution for monitoring data quality in supermarket grocery sales. By leveraging tools like IBM Watson Studio

and Cognos Analytics, organizations can dynamically visualize trends and anomalies, enabling proactive responses to data quality issues. The adoption of Isolation Forest ensures scalable and interpretable anomaly detection, while PCA aids in optimizing the feature set for enhanced model performance.

Outcomes:

- Enhanced automation in data quality monitoring.
- Real-time issue detection and resolution via IBM Cloud services.
- Interactive reporting and actionable insights to improve decision-making.

Next Steps

- Combine visualization insights with AI-driven anomaly detection to prioritize and address data quality issues.
- Deploy a scalable architecture leveraging **IBM Cloud**, incorporating AI for automated alerts and dashboards for real-time monitoring. Continuously refine data preparation techniques and AI models based on feedback and performance metrics.